

ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 8: Oscillator Phase Noise

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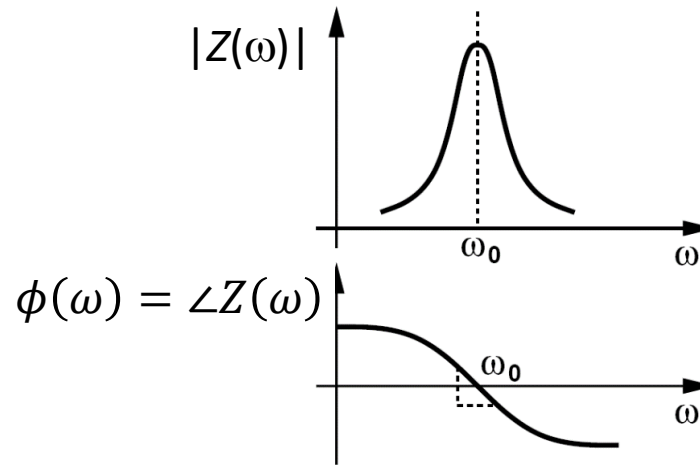
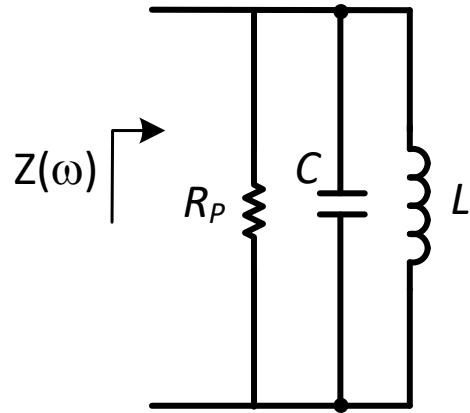


Outline

- Classical oscillator noise model – LTI (linear time invariant) model
 - Oscillator as a feedback system
 - General definition of Q
 - Leeson's equation
- Noise factor of LC oscillator
 - Review of device noise basics
 - Cyclostationary noise
 - Additive noise to phase noise conversion
- Bias noise impact on phase noise
- Phase noise in ring oscillators



Quality Factor Q



$$Q = \frac{\omega_0}{\omega_{3dB}}$$

$$Q = \frac{\omega_0}{2} \left| \frac{d\phi}{d\omega} \right|$$

Several definitions of Q

$$Q = \frac{R_p}{\omega L}$$

$$Q = R_p \omega C$$

$$Q = 2\pi \frac{\text{Energy Stored}}{\text{Energy Dissipated per Cycle}}$$

$$Z = \frac{1}{\frac{1}{R_p} + j\omega C + \frac{1}{j\omega L}} = \frac{1}{\frac{1}{R_p} + j\left(\omega C - \frac{1}{\omega L}\right)}$$

$$\angle Z = \phi = \tan^{-1} \left(R_p \left(\frac{1}{\omega L} - \omega C \right) \right)$$

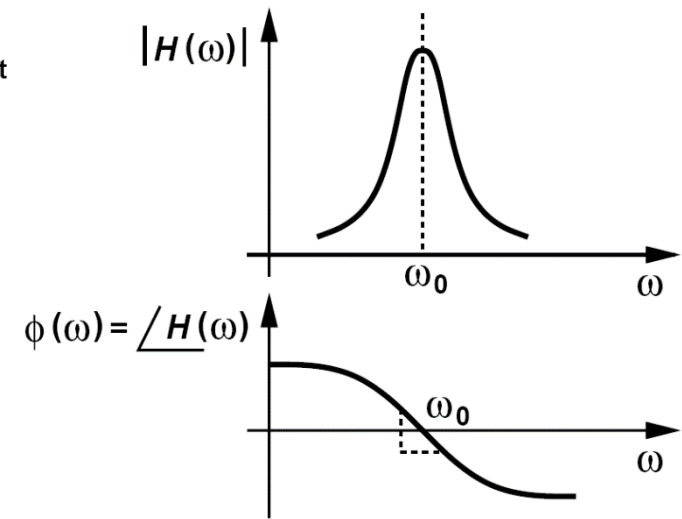
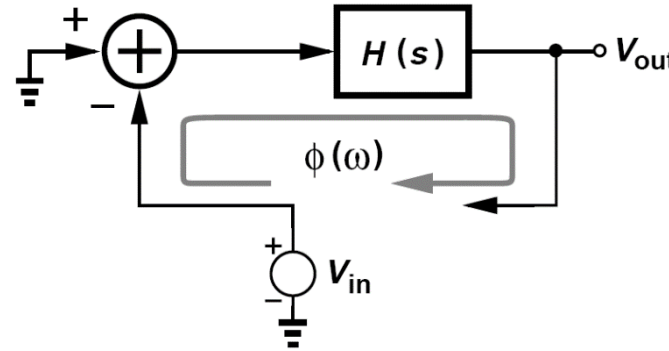
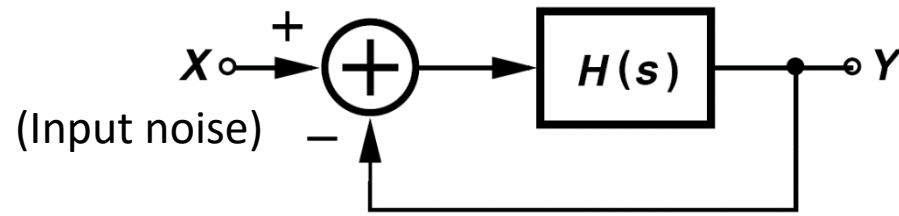
$$\frac{d\phi}{d\omega} = \frac{1}{1 + \left(R_p \left(\frac{1}{\omega L} - \omega C \right) \right)^2} \cdot \frac{d}{d\omega} \left(R_p \left(\frac{1}{\omega L} - \omega C \right) \right) = \frac{-R_p}{1 + \left(R_p \left(\frac{1}{\omega L} - \omega C \right) \right)^2} \left(\frac{1}{\omega^2 L} + C \right)$$

$$@ \omega = \omega_0 = \frac{1}{\sqrt{LC}}, \quad \frac{d\phi}{d\omega} = -\frac{2R_p}{\omega_0^2 L}$$

$$\text{If } Q = \frac{\omega_0}{2} \left| \frac{d\phi}{d\omega} \right|, \quad Q = \frac{R_p}{\omega_0 L}$$



Classical Oscillator Noise Model – LTI Model



From LC tank:

$$Q = \frac{R_P}{\omega_0 L}$$

$$Q = \frac{\omega_0}{2} \left| \frac{d\phi}{d\omega} \right|$$

$$H_{CL}(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + H(s)}$$

- Linear time invariant (LTI) model: oscillator as a feedback system with an open loop TF of $H(s)$ and close loop TF of $\frac{Y(s)}{X(s)} = \frac{H(s)}{1+H(s)}$ (1)
- Around the oscillation frequency ω_0 , $H(j\omega) \approx H(j\omega_0) + \Delta\omega \frac{dH}{d\omega}$ (2)
- $H(j\omega_0) \approx -1, \Delta\omega \frac{dH}{d\omega} \ll 1 \Rightarrow H_{CL}(j\omega) = \frac{Y}{X}(j\omega_0 + j\Delta\omega) \approx \frac{-1}{\Delta\omega \frac{dH}{d\omega}}$ (3)



Classical Oscillator Noise Model – LTI Model

From (3), $|H_{CL}(j\omega)|^2 = \left| \frac{Y}{X}(j\omega_0 + j\Delta\omega) \right|^2 \approx \left| \frac{-1}{\Delta\omega \frac{dH}{d\omega}} \right|^2$

Express $H(j\omega)$ in polar form, $H(j\omega) = |H|e^{j\phi}$

$$\frac{dH}{d\omega} = \frac{d|H|}{d\omega} e^{j\phi} + |H| \cdot j e^{j\phi} \frac{d\phi}{d\omega} \quad (4)$$

$$\left| \frac{dH}{d\omega} \right|^2 = \left| \frac{d|H|}{d\omega} \right|^2 + \left| \frac{d\phi}{d\omega} \right|^2 |H|^2 \quad (5)$$

$$\left| \frac{Y}{X}(j\omega_0 + j\Delta\omega) \right|^2 \approx \left| \frac{1}{\Delta\omega \frac{dH}{d\omega}} \right|^2 = \frac{1}{(\Delta\omega)^2} \cdot \frac{1}{\left| \frac{d|H|}{d\omega} \right|^2 + |H|^2 \left| \frac{d\phi}{d\omega} \right|^2} \quad (6)$$

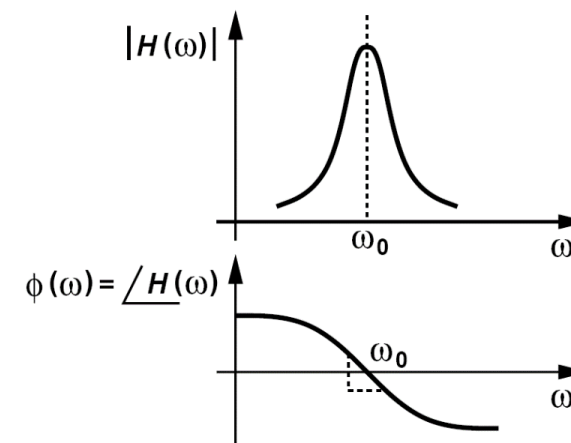
If we define Q around ω_0 as:

$$Q = \frac{\omega_0}{2} \frac{1}{\left| \frac{dH}{d\omega} \right|} = \frac{\omega_0}{2} \sqrt{\left| \frac{d|H|}{d\omega} \right|^2 + |H|^2 \left| \frac{d\phi}{d\omega} \right|^2} \quad (7)$$

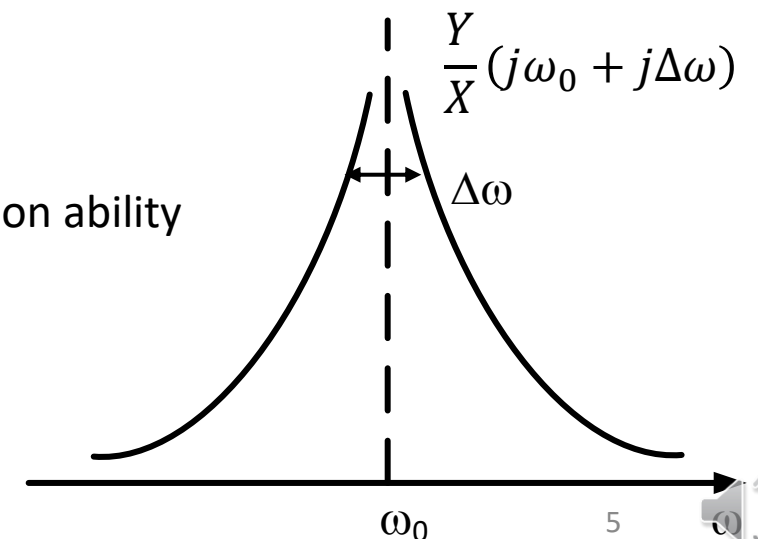
$$\left| \frac{Y}{X}(j\omega_0 + j\Delta\omega) \right|^2 \approx \left(\frac{\omega_0}{2\Delta\omega} \right)^2 \cdot \frac{1}{Q^2}, \text{ } Q \text{ indicates the FB system noise rejection ability}$$

Around ω_0 , $|H| \approx 1$, $\left| \frac{d|H|}{d\omega} \right| \approx 0$,

$$Q \approx \frac{\omega_0}{2} \left| \frac{d\phi}{d\omega} \right|, \quad |H_{CL}(j\omega)|^2 = \frac{1}{Q^2} \left(\frac{\omega_0}{2\Delta\omega} \right)^2 \quad (8)$$



$H_{CL}(j\omega_0 + j\Delta\omega)$



From LC tank:

$$Q = \frac{R_p}{\omega_0 L}$$

$$Q = \frac{\omega_0}{2} \left| \frac{d\phi}{d\omega} \right|$$

Leeson's Equation

- If input is the resistor thermal noise, $\frac{X(j\omega)}{\Delta f} = 4kTR$, and half of the noise power contributes to phase noise

$$\frac{|Y(j\omega + \Delta\omega)|^2}{\Delta f} = \frac{\overline{v_n^2}}{\Delta f} = 2kTR \cdot |H_{CL}(j\omega + \Delta\omega)|^2 = 2kTR \frac{1}{4Q^2} \left(\frac{\omega_0}{\Delta\omega}\right)^2 \quad (9)$$

- Leeson's equation (noise power to signal power ratio in dB):

$$L(\Delta\omega) = 10 \cdot \log \left(\frac{2kT}{P_{sig}} \frac{1}{4Q^2} \left(\frac{\omega_0}{\Delta\omega}\right)^2 \right) \quad (10) \quad \left(\text{Note: } \overline{p_n} = \frac{\overline{v_n^2}}{R} \right)$$

$$\text{VCO figure of merit: } FOM = 10 \log \left[\left(\frac{f_0}{\Delta f}\right)^2 \frac{1}{L(\Delta f) P_{DC, mW}} \right]$$

- Modifications from (10) to include

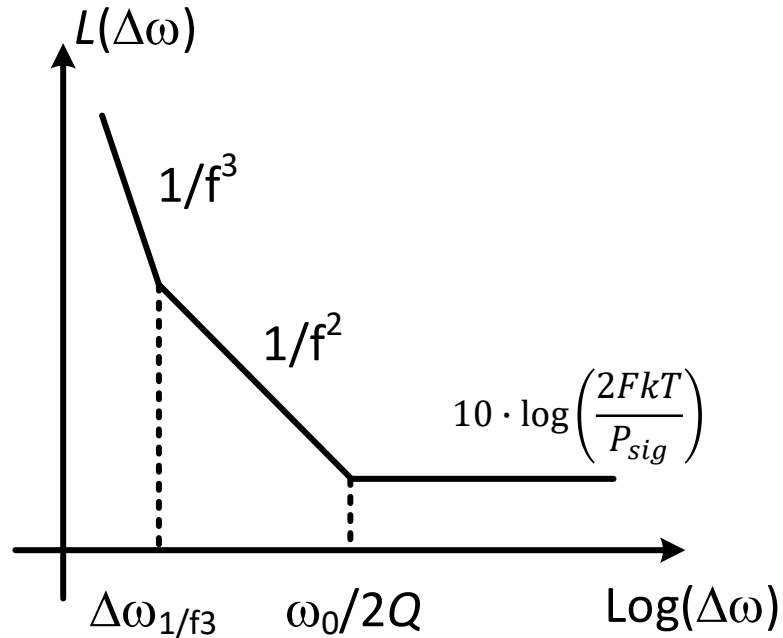
- Noise factor F including noise contribution from R_p , G_m and other sources
- $1/f^3$ region ($1/f$ noise)
- Flat region

$$L(\Delta\omega) = 10 \cdot \log \left(\frac{2FkT}{P_{sig}} \left(1 + \frac{1}{4Q^2} \left(\frac{\omega_0}{\Delta\omega}\right)^2 \right) \left(1 + \frac{\Delta\omega_{1/f^3}}{|\Delta\omega|} \right) \right) \quad (11)$$

Ref [2]



Phase Noise Profile



$$L(\Delta\omega) = 10 \cdot \log \left(\frac{2FkT}{P_{sig}} \left(1 + \frac{1}{4Q^2} \left(\frac{\omega_0}{\Delta\omega} \right)^2 \right) \left(1 + \frac{\Delta\omega_{1/f^3}}{|\Delta\omega|} \right) \right) \quad (12)$$

- Input white noise is shaped to $1/f^2$ noise by the feedback loop, 20dB/dec slope
- Input flicker noise $1/f$ is shaped to $1/f^3$, 30dB/dec
- Corner frequency $\Delta\omega_{1/f^3}$ is empirical and not exactly the same as device flicker noise corner
- $L(\Delta\omega) \propto \omega_0$, $\because \Delta\phi = \frac{\Delta t}{T_0} 2\pi = \Delta t \cdot \omega_0$, Δt is noise induced time jitter caused by noise Δv .
 - Phase noise increases 6dB when frequency doubles
- $L(\Delta\omega) \propto 1/Q$, high Q tank reduces phase noise

How to calculate noise factor F ?



Noise Factor of LC Oscillator

Basic device noise models

- Resistor thermal noise

- Voltage noise spectral density: $\frac{v_{n,R}^2}{\Delta f} = 4kTR$ (13)

- Current noise spectral density: $\frac{i_{n,R}^2}{\Delta f} = \frac{4kT}{R}$ (14)

- MOSFET thermal noise

- Voltage noise spectral density: $\frac{v_n^2}{\Delta f} = \frac{4kT\gamma}{g_m}$ (15)

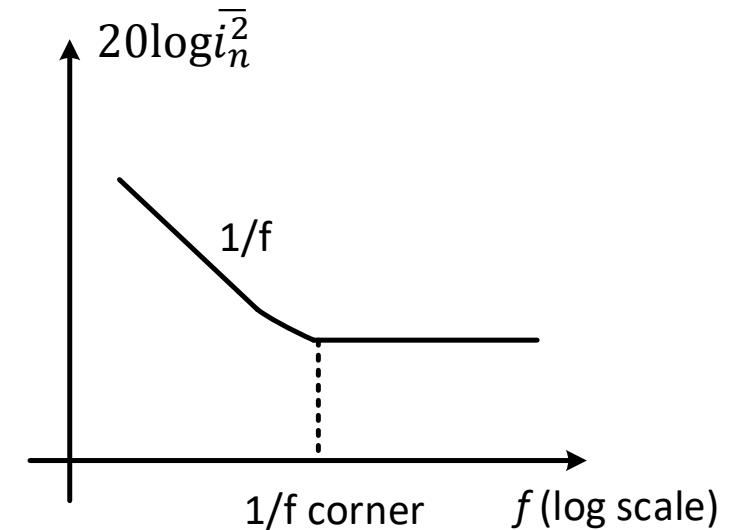
- Current noise spectral density: $\frac{i_n^2}{\Delta f} = 4kT\gamma g_m$ (16)

- $\gamma=2/3$ for long channel device and $2\sim3$ for short channel device

- MOSFET flicker noise

- $\frac{i_n^2}{\Delta f} = \frac{K}{f} \cdot \frac{g_m^2}{WLC_{ox}^2}$ (17)

- Larger device size ($W \cdot L$) reduces flicker noise



Noise Factor of LC Oscillators

Applying KCL at nodes X and Y:

$$I_{n1} + I_{D1} + I_{G2} = -I_X \quad (a)$$

$$I_{n2} + I_{D2} + I_{G1} = I_X \quad (b)$$

(a) - (b):

$$2I_X = (I_{n2} - I_{n1}) + (I_{D1} - I_{D2}) + (I_{G2} - I_{G1}) \quad (c)$$

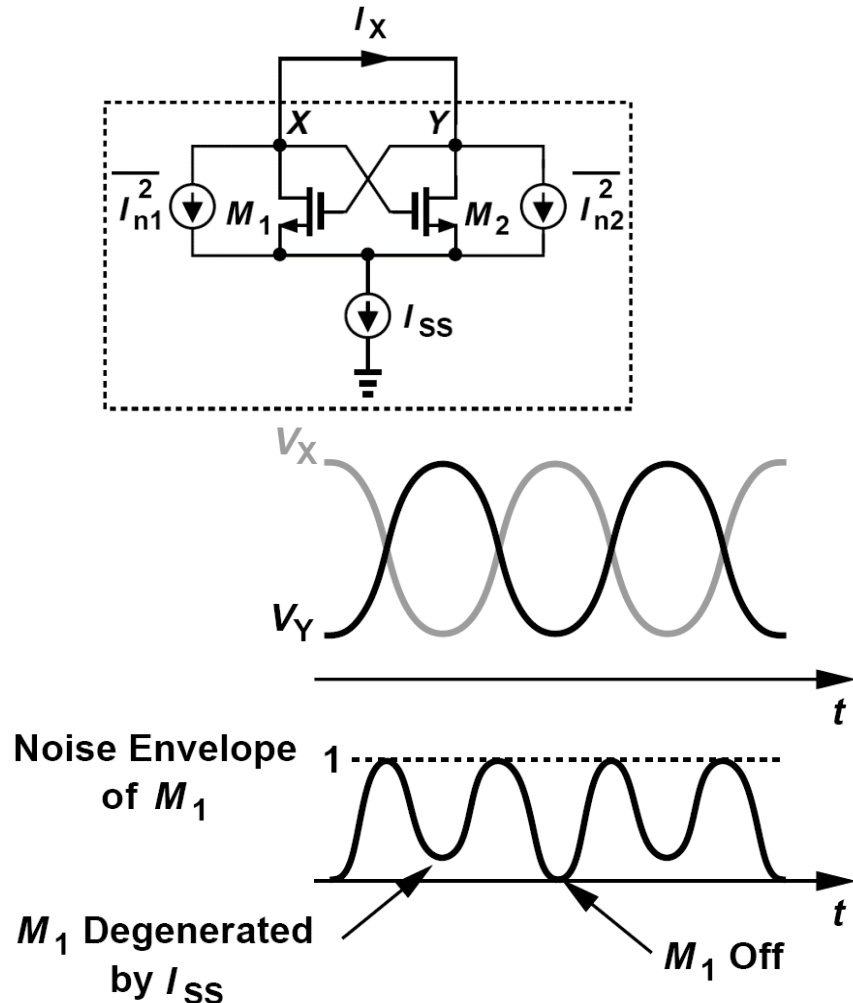
At equilibrium, $I_{D1} = I_{D2}$, $I_{G1} = I_{G2}$,

$$(c) \text{ Becomes } I_X = \frac{I_{n2} - I_{n1}}{2}.$$

$$\overline{I_X^2} = \overline{\left(\frac{I_{n2} - I_{n1}}{2}\right)^2} = \frac{\overline{I_{n1}^2} - 2\overline{I_{n1} \cdot I_{n2}} + \overline{I_{n2}^2}}{4} \quad (d)$$

$$\because I_{n1} \text{ and } I_{n2} \text{ are uncorrelated, } \overline{I_{n1} \cdot I_{n2}} = 0$$

$$\Rightarrow \overline{I_X^2} = \frac{\overline{I_{n1}^2} + \overline{I_{n2}^2}}{4} = kT\gamma g_m \quad (18)$$

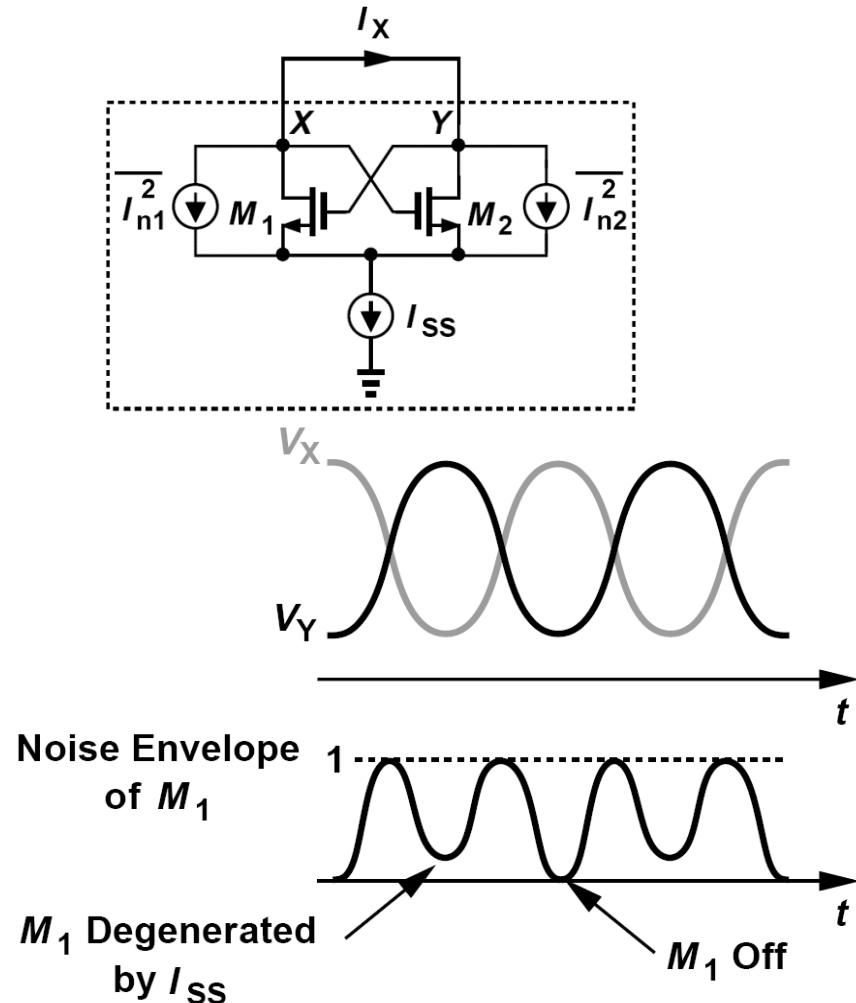


Ref [1]



Noise Factor of LC Oscillators

Calculate total cyclostationary noise



- G_m MOSFET noise at equilibrium:

$$I_X = \frac{I_{n2} - I_{n1}}{2}, \quad \overline{I_X^2} = \frac{\overline{I_{n1}^2} + \overline{I_{n2}^2}}{4} = kT\gamma g_m \quad (18)$$

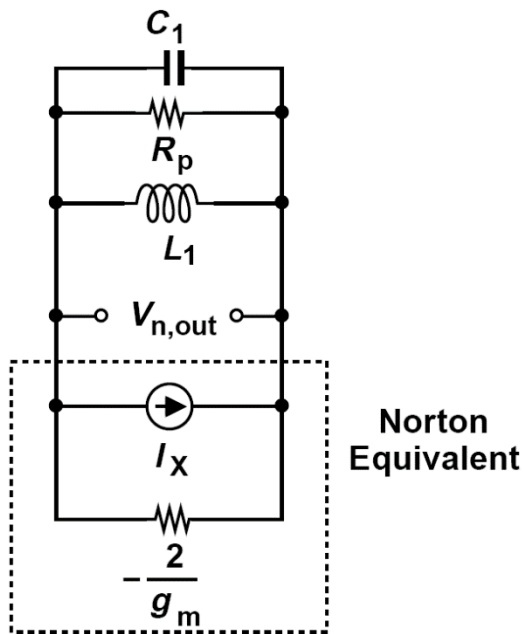
- G_m cyclostationary noise:

- V_X at maximum and V_Y at minimum: M_1 is off and does not inject noise to the tank
- V_X and V_Y at equilibrium: M_1 and M_2 both inject noise to the tank with a total spectral density of $kT\gamma g_m$.
- V_Y at maximum and V_X at minimum: M_1 is on and injects small amount of noise due to tail current degeneration.
- Model the injected noise as $kT\gamma g_m \left(\frac{1 + \cos(2\omega_0 t)}{2} \right)^2$ as an approximation, yielding an average spectral density of $\overline{I_{X,avg}^2} = \frac{3}{8} kT\gamma g_m$ (19)



Noise Factor of LC Oscillators

Calculate total cyclostationary noise



$$\text{Tank resistance noise: } I_{n,R_P}^2 = \frac{2kT}{R_P} \quad (20)$$

$$\text{Total tank current noise: } \overline{i_{n,tot}^2} = \frac{3}{8}kT\gamma g_m + \frac{2kT}{R_P} \quad (21)$$

$$\text{LC tank impedance } Z_{tank}^2 = \frac{R^2 L_1^2 \omega^2}{R^2 (1 - L_1 C_1 \omega^2)^2 + L_1^2 \omega^2} \quad (22)$$

$$R = -\frac{2}{g_m} // R_P$$

In cyclostationary steady state,

$$R = -\frac{1}{G_{avg}} // R_P \approx \infty, Z_{tank}^2 \approx \frac{L_1^2 \omega^2}{(1 - L_1 C_1 \omega^2)^2} \quad (23)$$

$$\text{For } \omega_0 = \frac{1}{\sqrt{L_1 C_1}}, \omega = \omega_0 + \Delta\omega, \Delta\omega \ll \omega_0, Z_{tank}^2 \approx \frac{R_P^2}{4Q^2} \left(\frac{\omega_0}{\Delta\omega} \right)^2 \quad (24)$$

Total additive cyclostationary noise:

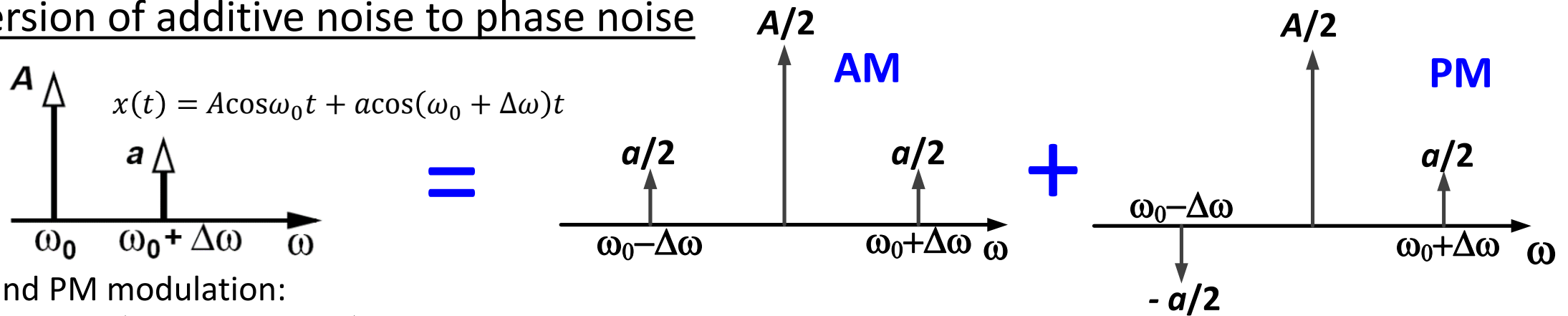
$$\overline{V_{n,out}^2} = \overline{i_{n,tot}^2} \cdot Z_{tank}^2, \overline{V_{n,out}^2} = kT \left(\frac{3}{8}\gamma g_m + \frac{2}{R_P} \right) \frac{R_P^2}{4Q^2} \left(\frac{\omega_0}{\Delta\omega} \right)^2 \quad (25)$$

Ref [1]



Noise Factor of LC Oscillators

Conversion of additive noise to phase noise



AM and PM modulation:

$$x_{AM}(t) = \frac{A}{2} \left(1 + \frac{2a}{A} \cos\Delta\omega t \right) \cos\omega_0 t = \frac{A}{2} \cos\omega_0 t + \frac{a}{2} \cos(\omega_0 - \Delta\omega)t + \frac{a}{2} \cos(\omega_0 + \Delta\omega)t \quad (26)$$

$$x_{PM}(t) = \frac{A}{2} \cos \left(\omega_0 t + \frac{2a}{A} \sin\Delta\omega t \right) \approx \frac{A}{2} \cos\omega_0 t - a \sin\omega_0 t \cdot \sin\Delta\omega t$$

$$= \frac{A}{2} \cos\omega_0 t - \frac{a}{2} \cos(\omega_0 - \Delta\omega)t + \frac{a}{2} \cos(\omega_0 + \Delta\omega)t \quad (27)$$

Decompose sideband to AM and PM noise:

$$x(t) = A\cos\omega_0 t + a\cos(\omega_0 + \Delta\omega)t$$

$$= \underbrace{\frac{A}{2} \cos\omega_0 t + \frac{a}{2} \cos(\omega_0 - \Delta\omega)t + \frac{a}{2} \cos(\omega_0 + \Delta\omega)t}_{\text{AM noise}} + \underbrace{\frac{A}{2} \cos\omega_0 t - \frac{a}{2} \cos(\omega_0 - \Delta\omega)t + \frac{a}{2} \cos(\omega_0 + \Delta\omega)t}_{\text{PM noise}} \quad (28)$$



Noise Factor of LC Oscillators

Calculation of phase noise spectrum and noise factor F

AM noise is removed by a hard limiter (e.g. buffers) that clips the amplitude.

The output of the limiter contains PM noise:

$$x_{lim}(t) = \frac{A}{2} \cos \omega_0 t - \frac{a}{2} \cos(\omega_0 - \Delta\omega)t + \frac{a}{2} \cos(\omega_0 + \Delta\omega)t \approx \frac{A}{2} \cos \left(\omega_0 t + \frac{2a}{A} \sin \Delta\omega t \right) \quad (29)$$

Compare the additive noise $x(\Delta\omega)$ and the phase noise $S_\phi(\Delta\omega)$

- The additive noise: $x(\Delta\omega) = a \cos(\omega_0 + \Delta\omega)t$ (30)

- The PM component: $\frac{2a}{A} \sin \Delta\omega t$ (31)

In the previous derivation, the additive noise is $\overline{V_{n,out}^2} = kT \left(\frac{3}{8} \gamma g_m + \frac{2}{R_P} \right) \frac{R_P^2}{4Q^2} \left(\frac{\omega_0}{\Delta\omega} \right)^2$ (25)

Phase noise spectrum $S_\phi(\Delta\omega) = \frac{\overline{V_{n,out}^2}}{A^2/2} = \frac{2}{A^2} kT \left(\frac{3}{8} \gamma g_m + \frac{2}{R_P} \right) \frac{R_P^2}{4Q^2} \left(\frac{\omega_0}{\Delta\omega} \right)^2$ (32)

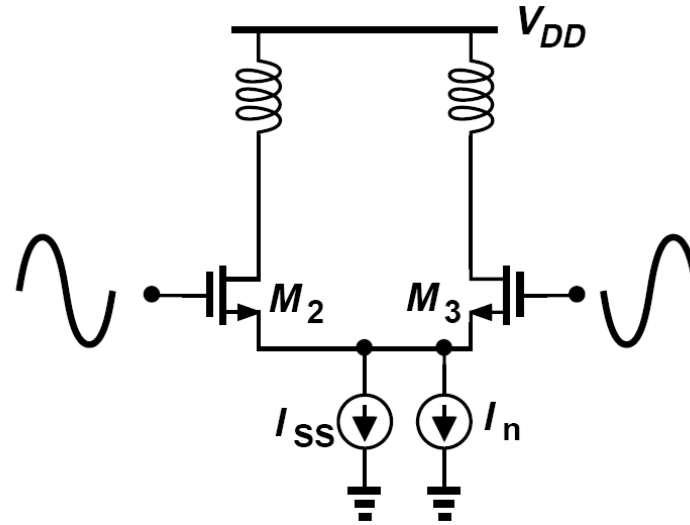
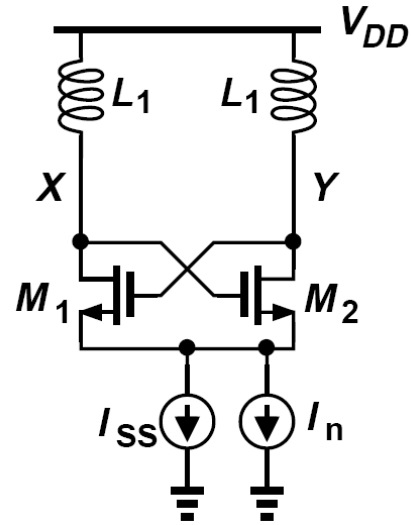
On the other hand, Leeson's equation: $L(\Delta\omega) = 10 \cdot \log \left(\frac{2FkT}{P_{sig}} \frac{1}{4Q^2} \left(\frac{\omega_0}{\Delta\omega} \right)^2 \right)$, where $P_{sig} = \frac{A^2}{2R_P}$

$\Rightarrow$ Noise factor $F = \frac{1}{2} R_P \left(\frac{3}{8} \gamma g_m + \frac{2}{R_P} \right)$ (33)

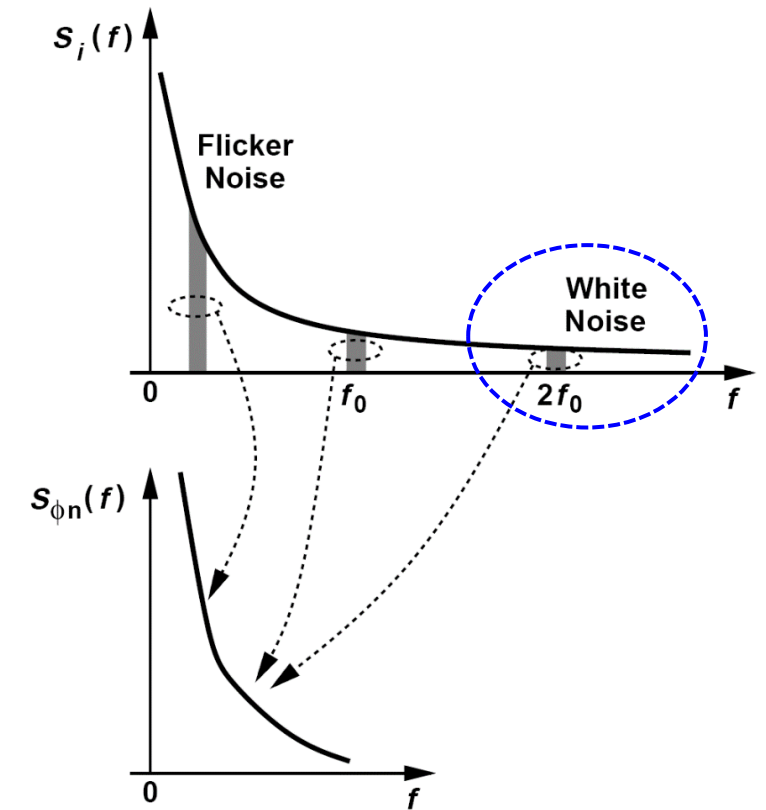
$g_m = \frac{2}{R_P}, F = \frac{3}{8} \gamma + 1$ (34)



Bias Current Noise Impact on Phase Noise



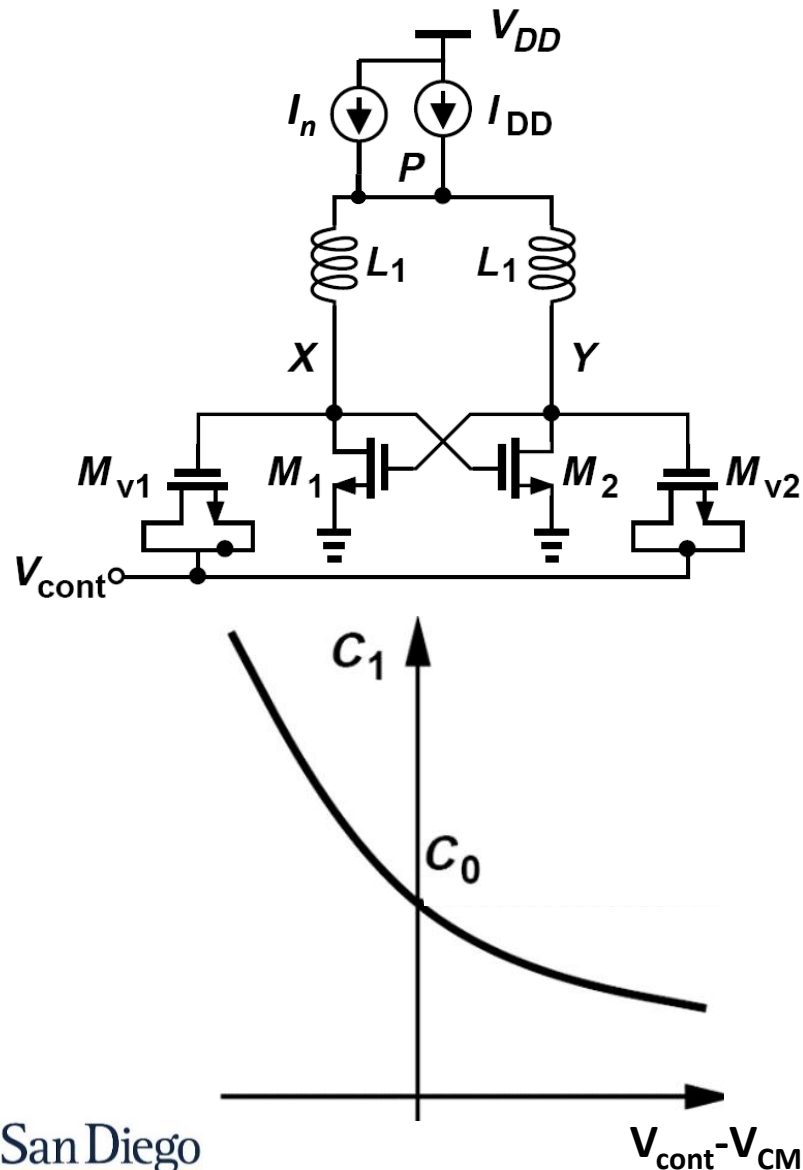
- LC oscillator as a mixer with an LO signal at ω_0 .
- Tail current thermal noise around $2\omega_0$ is down converted to ω_0 .
- The thermal noise around $4\omega_0$, $6\omega_0$, ..., will also be down converted by the harmonics of the LO to ω_0 .



Ref [1]



Bias Current Noise Impact on Phase Noise

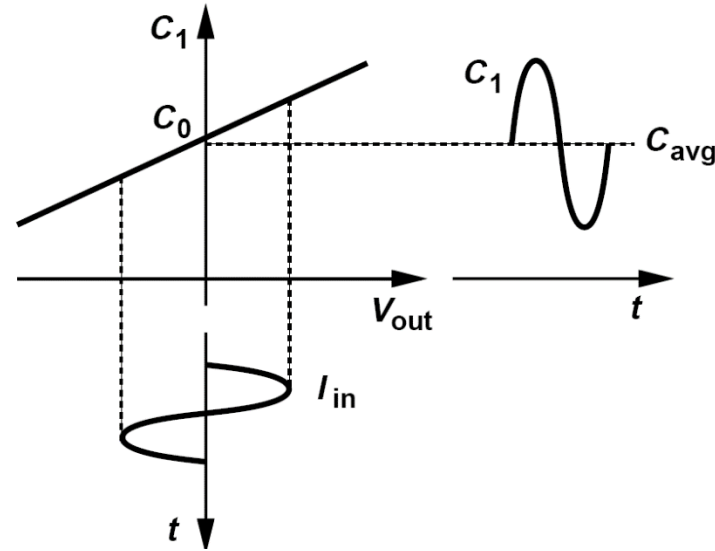
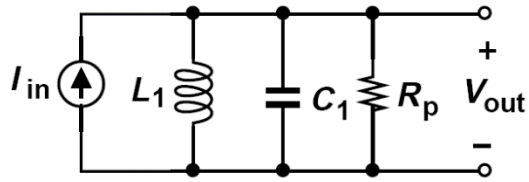


- Bias current noise I_n modulates VCO tank common mode voltage (V_{CM} of nodes X and Y)
- V_X and V_Y control varactor capacitance as V_{cont} .
- Therefore bias current noise I_n modulates the oscillator frequency and generates phase noise
- To avoid common mode voltage impact on varactor capacitance, use AC coupled connection. Need noise filtering on varactor bias voltage (very low filtering bandwidth).

Ref [1]



AM-PM Noise Conversion



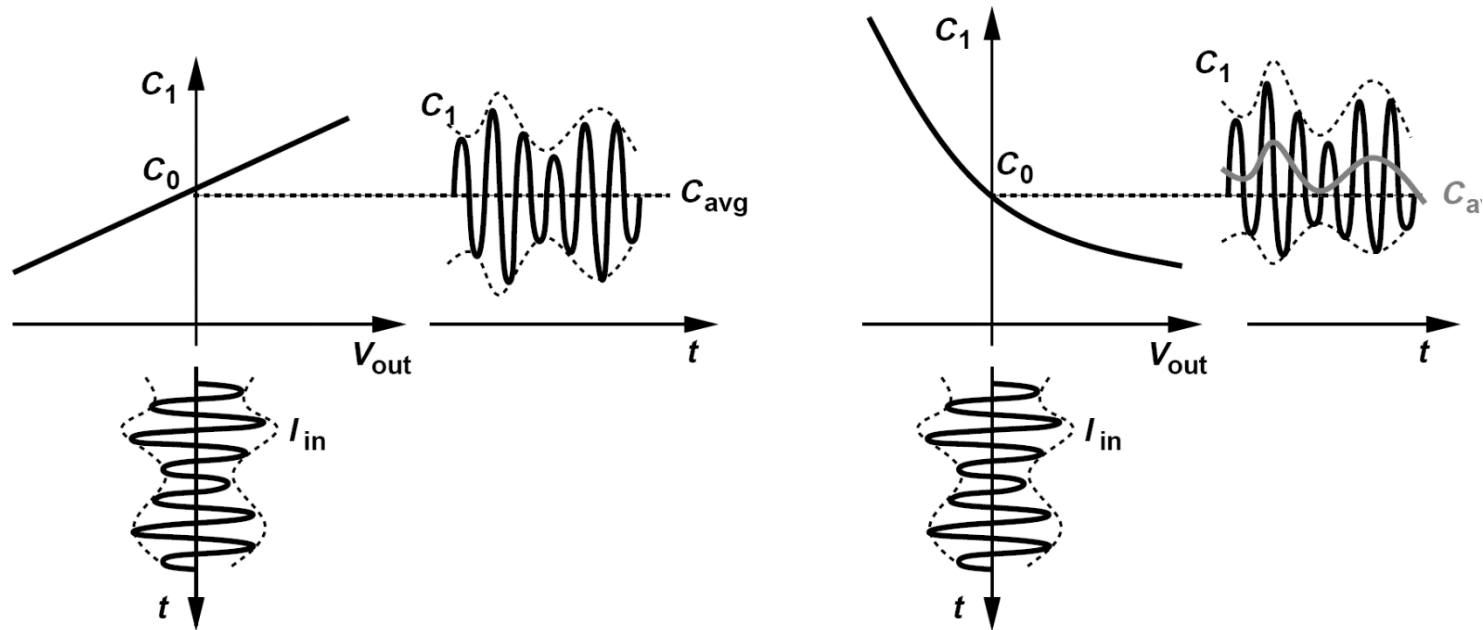
$$C_{var} = C_0 + f(V_{tank}) = C_0 + \alpha \cdot V_0 \sin \omega t$$

- Varactor capacitance is a function of tank voltage, approximately sinusoidal.
- If varactor C-V characteristic is linear, varactor capacitance is also sinusoidal. The average of C_{var} is C_0 , independent of V_0 .

Ref [1]



AM-PM Noise Conversion



- If C-V has nonlinear voltage dependence, e.g., $C_{var} = C_0 + \alpha_1 \cdot V + \alpha_2 \cdot V^2$
- Capacitance changes more sharply above C_{avg} and less below C_{avg} . A larger amplitude yields a higher C_{avg} .
- Bias current modulates tank amplitudes and average capacitance, thus changes oscillation frequency and contributes phase noise. Amplitude noise to phase noise conversion.
- Sources of nonlinear capacitance: varactor, Gm device junction capacitance.

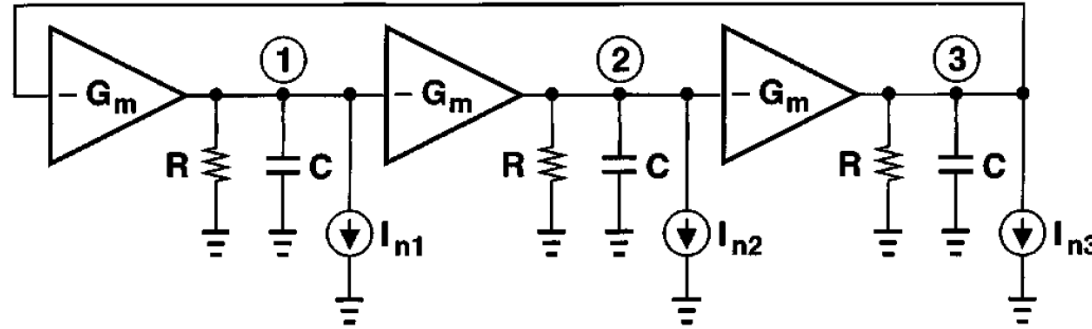
Ref [1]



Phase Noise of Ring Oscillators

$$H(s) = - \left(\frac{A_0}{1 + j \frac{\omega}{\omega_{RC}}} \right)^N$$

$$H_{CL}(j\omega) = \frac{Y}{X} (j\omega_0 + j\Delta\omega) \approx \frac{-1}{\Delta\omega \frac{dH}{d\omega}}$$



Naming convention:
 ω_0 : oscillation freq.
 ω_{RC} : RC 3dB freq.

- 3-stage ring oscillator

- Gain of each stage: $A_0 = G_m \cdot R = \sqrt{1 + \tan^2 \left(\frac{180}{N} \right)} = 2$ (35)

- Oscillation frequency: $\omega_0 = \omega_{RC} \tan \left(\frac{180}{N} \right) = \frac{1}{RC} \tan \left(\frac{180}{N} \right) = \frac{\sqrt{3}}{RC}$ (36)

- Open loop transfer function: $H(j\omega) = \frac{-1}{\left(1 + j\sqrt{3} \frac{\omega}{\omega_0} \right)^3}$ (37)

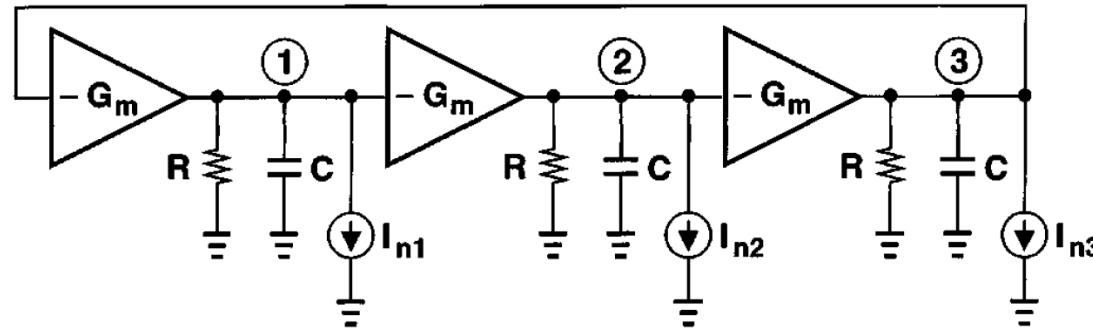
- $\left| \frac{V_1}{I_{n1}} (j\omega_0 + j\Delta\omega) \right|^2 \approx \left| \frac{1}{\Delta\omega \frac{dH}{d\omega}} \right|^2 \cdot \left| R // \frac{1}{j\omega C} \right|^2 \Big|_{\omega_0} = \frac{1}{(\Delta\omega)^2} \cdot \frac{1}{\left| \frac{d|H|}{d\omega} \right|^2 + \left| \frac{d\phi}{d\omega} \right|^2} \cdot \left| R // \frac{1}{j\omega C} \right|^2 \Big|_{\omega_0}$
- For $\left| \frac{d|H|}{d\omega} \right|_{\omega_0} = \frac{9}{4\omega_0}$, $\left| \frac{d\phi}{d\omega} \right|_{\omega_0} = \frac{3\sqrt{3}}{4\omega_0}$, $\left| \frac{V_1}{I_{n1}} (j\omega_0 + j\Delta\omega) \right|^2 = \frac{R^2}{27} \left(\frac{\omega_0}{\Delta\omega} \right)^2$ (38)

- Feedback system Q: $Q = \frac{\omega_0}{2} \sqrt{\left| \frac{d|H|}{d\omega} \right|^2 + \left| \frac{d\phi}{d\omega} \right|^2} = \frac{3\sqrt{3}}{4} \approx 1.3$ (39)



Phase Noise of Ring Oscillators

Additive Noise



$$\left| \frac{V_1}{I_{n1}} (j\omega_0 + j\Delta\omega) \right|^2 \approx \frac{R^2}{27} \left(\frac{\omega_0}{\Delta\omega} \right)^2 \quad (39)$$

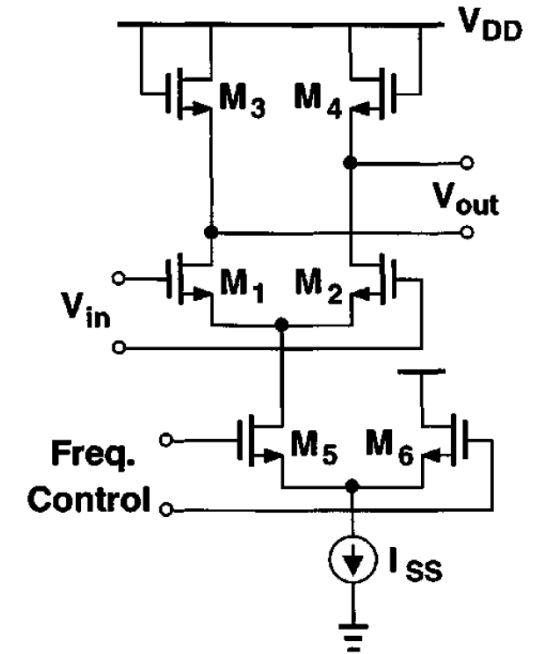
$$\text{Additive noise, assume } \overline{I_{n1}^2} = \overline{I_{n2}^2} = \overline{I_{n3}^2} = \overline{I_n^2}$$

$$\text{Total output noise power is } |V_{1tot}(j\omega_0 + j\Delta\omega)|^2 = \frac{R^2}{9} \left(\frac{\omega_0}{\Delta\omega} \right)^2 \overline{I_n^2} \quad (40)$$

$$\overline{I_n^2} = 4kT\gamma(g_{m1} + g_{m2}) \approx \frac{8kT\gamma}{R}, R = \frac{1}{g_{m3}} = \frac{1}{g_{m4}} \approx \frac{1}{g_{m1}}$$

$$|V_{1tot}(j\omega_0 + j\Delta\omega)|^2 = \frac{8kT\gamma R}{9} \left(\frac{\omega_0}{\Delta\omega} \right)^2 \quad (41)$$

$$\text{Additive noise} \rightarrow \text{phase noise: } S_\phi(\Delta\omega) = \frac{\overline{V_{1tot}^2}}{A^2/2} = \frac{2}{A^2} \frac{8kT\gamma R}{9} \left(\frac{\omega_0}{\Delta\omega} \right)^2 \quad (42)$$

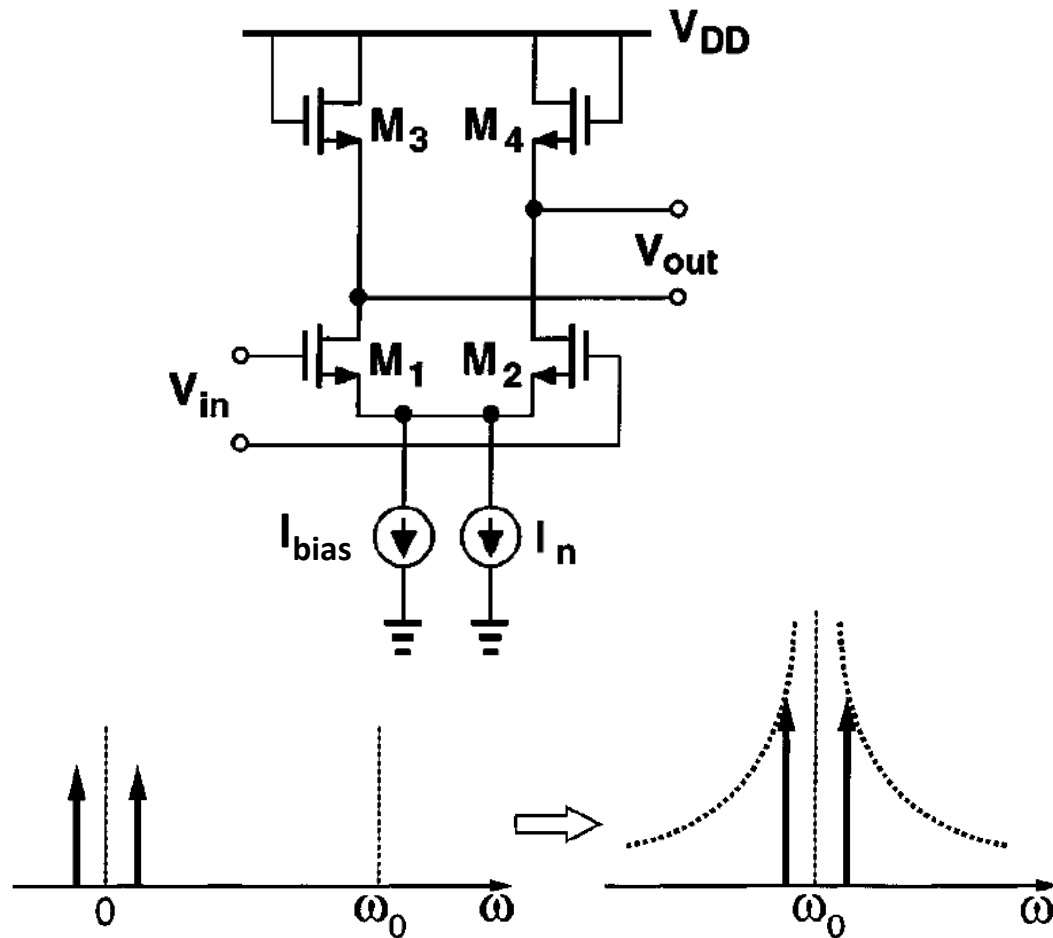


Ref [3]



Phase Noise in Ring Oscillators

Low Frequency Multiplicative Noise



- Oscillation frequency tuned by tail current I_{bias} .
- Noise in I_{SS} modulates the frequency and contributes to phase noise. Not predicted by LTI model as the additive noise.

- VCO sensitivity to I_{bias} : $K_{Ibias} = \frac{d\omega_{osc}}{dI_{bias}}$, $I_n = I_m \cos \omega_m t$

$$\omega = \omega_0 + K_{Ibias} \cdot I_m \cos \omega_m t$$

$$V_{out}(t) = V_0 \cos \left(\omega_0 t + K_{Ibias} \int I_m \cos \omega_m t dt \right) =$$

$$V_0 \cos \left(\omega_0 t + K_{Ibias} \frac{I_m}{\omega_m} \sin \omega_m t \right) \approx$$

$$V_0 \cos \omega_0 t + \frac{V_0 K_{Ibias} I_m}{2\omega_m} [\cos(\omega_0 + \omega_m)t - \cos(\omega_0 - \omega_m)t]$$

- The ratio of each noise sideband amplitude to the carrier amplitude:

$$\left| \frac{V_n}{V_0} \right|^2 = \frac{1}{4} \left(\frac{K_{Ibias}}{\omega_m} \right)^2 I_m^2$$

Ref [3]



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